

The effect of blood removal on oxidation and shelf life of broiler breast meat.

Blood components, especially hemoglobin, are powerful promoters of lipid oxidation and may decrease the shelf life of meat products. Therefore, this study examined different slaughter techniques to determine their effects on pH (24 h), color ($L^*a^*b^*$ values at 24 h), lipid oxidation, residual hemoglobin concentration (24 h), and sensory evaluation (d 1 and 4 postmortem; PM) in broiler breast fillets. The treatments included 1) CO₂ slaughter and not bled, 2) no stunning and bled, 3) electrical stunning (ES) and bled, 4) CO₂ stunning and bled, and 5) ES and decapitation. The birds were conventionally processed, and analyses were performed at 24 h PM except residual hemoglobin for which the samples were frozen (-80 degrees C) until analyses (< 2 mo). There were no significant differences in pH or b^* values at 24 h PM among any of the treatments. L^* values were significantly higher, indicating lighter fillets in the ES and decapitated birds compared with the darker fillets from the CO₂ stunned and bled birds. The CO₂ slaughter and not bled birds had significantly higher a^* values, indicating more red color, when compared with the ES and bled and decapitated birds. There were no significant differences in the residual hemoglobin contents in the broiler breast muscle when comparing all of the treatments except CO₂ slaughter and not bled, which was significantly (around 15%) greater. Overall TBA-reactive substances (TBARS; raw, cooked at 24 h, and cooked at 72 h PM) indicated that ES and bled birds had the lowest TBARS when compared with the remaining treatments. Consumer panels detected increased aroma (chicken meaty and warmed-over aromas) and flavor (chicken meaty and warmed-over flavors) in not bled samples at 24 h PM. By 72 h PM, however, there were no significant differences in aroma or flavor. Therefore, different slaughter and bleeding method may affect color and sensory properties of the broiler breast fillets, and the ES and decapitation method had the most favorable results for sensory quality.